Christopher S. Meiklejohn

Distributed systems researcher and technical lead who turns research into production systems

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CAREER HIGHLIGHTS

DoorDash - Staff Software Engineer (full-time) *Aug 2022 - present*

- Technical lead and co-designer of the unified ordering platform for DoorDash, Wolt, and Deliveroo, targeting multi-region, active-active operation.
- Designed a framework adopted by multiple teams for product features and observability; built migration validation for ordering data.

ACADEMIC APPOINTMENT & DOCTORAL RESEARCH

Carnegie Mellon University - Adjunct Faculty, Software and Societal Systems (Mar 2025 - present)

Teach CMU 17-313 guest lectures on safe deployment, microservice resilience, and AI-assisted software development.

Carnegie Mellon University, Ph.D. in Software Engineering (2024); advisor: Heather Miller.

Dissertation: *Resilient Microservice Applications, by Design, and without the Chaos*.

Doctoral research: systematic testing and fault injection to expose microservice resilience bugs before production.

SELECTED RESEARCH CONTRIBUTIONS & RECOGNITION

Lasp: PPDP 2025 10 Year Most Influential Paper Award. Co-created the coordination-free programming model with Peter Van Roy and implemented its Erlang prototype; scaled it to 1,024 Erlang nodes on AWS while contracting at Mesosphere. Original paper: PPDP 2015.

Filibuster: Service-Level Fault Injection Testing (SoCC 2021). Created Filibuster; brought resilience testing to DoorDash's DashPass with Kotlin/gRPC concurrency support. Co-authored circuit-breaker research (SoCC 2022) and execution indexing (2022 preprint); mentored database resilience research (ICSE Companion 2024).

PARTISAN: Scaling the Distributed Actor Runtime (USENIX ATC 2019). Built an Erlang distribution layer to scale Lasp. LeapSight deployed it in LoJack LATAM's vehicle-recovery platform: 300,000 vehicles and 30M+ GPS transmissions/day (2022); also adopted by BarrelDB.

RESEARCH INTO PRODUCTION & INDUSTRY COLLABORATION

Microsoft Research: Azure Durable Entities and Orleans. Co-developed the model and prototype behind Azure Durable Entities; co-implemented transactions shipped in Orleans. Co-authored Durable Functions (OOPSLA 2021) and both Netherite papers (PVLDB 2022; VLDB Journal 2025). AWS Lambda also offers durable execution.

Amazon Web Services: PObserve. Developed the foundation of PObserve, a distributed-systems runtime monitor still used across AWS to detect bugs missed by testing.

Basho / Riak and Machine Zone. Co-authored Riak DT Map; supported Riot Games' League of Legends in production and later consulted for the NHS. Built QuickCheck models under John Hughes. At Machine Zone, designed and co-implemented chain replication; analyzed correctness with Maurice Herlihy.

Research consulting: Macrometa and Veritas. Macrometa Consulting Scientist: formal analysis of C8DB replication and transactions, presented at HPTS 2019. Built Veritas's distributed-database communication layer from Ph.D. research.

Academic-industry collaboration and research funding. SyncFree research with Basho, Trifork, Rovio, and Erlang Solutions; contributor to the successful EUR 3.57M LightKone edge-computing proposal.

EMPLOYMENT

DoorDash - Staff Software Engineer (full-time) *Aug 2022 - present*

- Technical lead and co-designer of the unified ordering platform for DoorDash, Wolt, and Deliveroo, targeting multi-region, active-active operation.
- Designed an aspect-oriented framework adopted by multiple teams to add features and monitor behavior by product feature and business vertical.
- Designed and implemented dual writes and field mappings to migrate selected ordering data from legacy systems to the new platform.
- Built end-to-end checkout tests, shadow checks, and discrepancy detection to validate ordering migrations.

Machine Zone - Senior Software Engineer, Core Platform *2015 - 2016*

Designed and co-implemented a chain replication protocol in Erlang; analyzed its correctness with Maurice Herlihy using formal verification.

Basho Technologies - Senior Software Engineer *2012 - 2015*

Maintained Riak's database internals, multi-datacenter replication, and convergent data types; researched CRDTs as part of the SyncFree project. Built stateful QuickCheck models under John Hughes's mentorship. Supported Riot Games' League of Legends in production.

Swipely - Software Engineer *2010 - 2012*

Developed the Rails platform and purchase-history newsfeed; led the migration from Rails 2 to Rails 3.

Tapchamp - Lead Software Developer *2010*

Built a Rails fantasy-play prediction platform with Facebook integration and an iPhone API; managed its servers.

Berklee College of Music - Software Engineering Manager and Systems Engineer *2008 - 2010*

Led Berklee Media software development; maintained two data centers and provisioned a disaster-recovery site.

Artists House Music - Web Programmer and Systems Administrator *2008 - 2009*

Built custom marketing tools and provided staff support and systems administration.

Berklee College of Music - Software Developer and Systems Engineer *2006 - 2008*

Built and supported AOLServer/OpenACS applications and provided 24/7 support for web services.

Conversent Communications - Senior Systems Engineer *2005 - 2006*

Built high-availability email and web systems, multi-server LDAP infrastructure, and automated service provisioning.

Conversent Communications - Systems Engineer *2000 - 2005*

Deployed Linux, Solaris, Windows, and FreeBSD systems; built web tools for customer management.

IDS World Network - Technical Support / Customer Service *1998 - 2000*

Provided technical support and customer service by telephone, email, and newsgroups.

CONSULTING & CONTRACT WORK

DoorDash - Contractor *2021 - 2022*

Tested DashPass with Filibuster to find resilience bugs; co-authored Method Overloading the Circuit (SoCC 2022).

LeapSight - Contractor *2020 - 2021*

Integrated Partisan into Riak Core and designed reliable workflows, adapting ideas from earlier Durable Functions and Orleans research.

CONSULTING & CONTRACT WORK (CONTINUED)

Macrometa - Consulting Scientist (contractor) *2019 - 2021*

Analyzed C8DB's geo-replication and transaction semantics using formal verification; documented its algorithms, CRDT merging, and garbage collection. Built property-based failover tests; presented the joint work at HPTS 2019.

Helium - Consultant, Distributed Systems *Apr - Aug 2019*

Prototyped fault-injection testing for a HoneyBadgerBFT worker in Helium's blockchain stack.

UK National Health Service - Contractor *2018 - 2019*

Updated Riak's Erlang/OTP 20 compatibility and release tooling; addressed dependency warnings.

Adobe - Consultant, Core Tech / Manticore *2018*

Reviewed causal consistency and conflict-free replicated data type (CRDT) designs; proposed improvements to replication algorithms.

Veritas - Consultant, Senior Systems Engineer *2017 - 2018*

Applied Ph.D. research to build a communication layer for a distributed database; developed property-based tests for network partitions and quorum reads/writes.

Comcast - Consultant, Distributed Systems *2016*

Designed and co-implemented a scalable, fault-tolerant membership service.

Input Output HK (IOHK) - Consultant, Cloud Haskell / OTP *2016*

Advised on Erlang/OTP-style abstractions for distributed programming in Cloud Haskell.

Mesosphere - Independent Contractor, Distributed Systems *2016*

Built high-availability DNS for DC/OS; reviewed the architecture and code of its high-availability load balancer.

South End Community Health Center - Freelance Web Developer *2010*

Built English and Spanish versions of a website in collaboration with a designer.

Coevolution Interactive - Freelance Web Developer *2009 - 2010*

Provided e-commerce and web development for clients.

Crunchy Granola Baby - Freelance Developer *2008 - 2009*

Designed and implemented the store's e-commerce site.

ACADEMIC APPOINTMENT & EDUCATION

Carnegie Mellon University - Adjunct Faculty, Software and Societal Systems *Mar 2025 - present*

Teach guest lectures on software engineering, microservice reliability, and AI-assisted development.

Carnegie Mellon University - Ph.D. in Software Engineering (2024)

Dissertation: *Resilient Microservice Applications, by Design, and without the Chaos*
Advisor: Heather Miller. Pittsburgh, PA.
Doctoral research: Develops systematic fault injection and principled testing to detect microservice resilience bugs before production.

Université catholique de Louvain - Doctoral study in Computer Science (Transferred 2018)

Joint program with Instituto Superior Técnico, Lisbon; advisors: Peter Van Roy and Rodrigo Rodrigues.
Louvain-la-Neuve, Belgium; transferred to Carnegie Mellon in August 2018.
Erasmus Mundus Joint Doctorate Fellowship (2015); MOBILITY Fellowship (2016).

Northeastern University - B.S. in Information Technology (2009)

Boston, MA. Summa Cum Laude; Sigma Epsilon Rho Honor Society; GPA: 3.75/4.0.
Academic Excellence Award (2009).

Community College of Rhode Island - A.S. in Computer Science (Warwick, RI) (2006)

RESEARCH INTERNSHIPS

DoorDash - Intern *Summer 2021*

Mentor: Matt Ranney. Adapted Filibuster to DoorDash's Kotlin/gRPC services, extending fault-injection algorithms for concurrency and enabling runtime instrumentation without application-code changes.

Co-authored: Service-Level Fault Injection Testing (SoCC 2021)

Amazon Web Services - Research Intern *Summer 2020*

Mentor: Ankush Desai. Developed the foundation of PObserve, a runtime monitor for distributed systems still used across AWS to detect bugs missed by testing.

Project discussed by Brooker and Desai: Systems Correctness Practices at AWS (CACM 2025)

Microsoft Research - Research Intern *Summer 2019*

Mentor: Sebastian Burckhardt. Researched postmortem debugging and deterministic replay with algebraic effects.

Co-authored related platform paper: A.M.B.R.O.S.I.A. (PVLDB 2020)

Microsoft Research - Research Intern *Summer 2018*

Mentor: Sebastian Burckhardt. Researched stateful serverless programming; the resulting capabilities shipped as Azure Durable Entities.

Co-authored: Durable Functions (OOPSLA 2021); Netherite (PVLDB 2022); Netherite (VLDB J. 2025, extended journal article)

Microsoft Research - Research Intern *Summer 2017*

Mentor: Phil Bernstein. Co-implemented distributed transactions that shipped in Orleans.

Acknowledged contributor: Cloud Actor-Oriented Database Transactions in Orleans (PVLDB 2024)

PUBLICATIONS

Dissertation

1. **Resilient Microservice Applications, by Design, and without the Chaos**

Christopher S. Meiklejohn. *Sole author.* Ph.D. dissertation, Software and Societal Systems, Carnegie Mellon University (2024); CMU-S3D-24-104.

Develops systematic fault injection and principled testing to detect microservice resilience bugs before production.

Journal and Conference Papers

2. **Netherite: Efficient Execution of Serverless Workflows**

Sebastian Burckhardt, Badrish Chandramouli, Chris Gillum, David Justo, Konstantinos Kallas, Connor McMahon, **Christopher S. Meiklejohn**, Xiangfeng Zhu. The VLDB Journal 34, article 25 (2025). *Extended journal version of [5].*

Contribution: Co-developed the stateful-serverless model and ReactiveMachine prototype preceding Netherite.

3. **Distributed, Coordination-Free Programming: 10 Years of Progress Since Lasp**

Christopher Meiklejohn, Peter Van Roy. PPDP 2025, pp. 1-3. Invited retrospective for the PPDP 2025 10 Year Most Influential Paper Award.

4. **Method Overloading the Circuit**

Christopher Meiklejohn, Lydia Stark, Cesare Celozzi, Matt Ranney, Heather Miller. SoCC 2022, pp. 273-288.

Contribution: Applied Filibuster to DashPass; documented DoorDash's circuit-breaker implementation and failure cases.

5. **Netherite: Efficient Execution of Serverless Workflows**

Sebastian Burckhardt, Badrish Chandramouli, Chris Gillum, David Justo, Konstantinos Kallas, Connor McMahon, **Christopher S. Meiklejohn**, Xiangfeng Zhu. Proceedings of the VLDB Endowment 15(8), pp. 1591-1604 (2022).

Contribution: See the shared programming-model contribution in [2].

6. **Service-Level Fault Injection Testing**

Christopher S. Meiklejohn, Andrea Estrada, Yiwen Song, Heather Miller, Rohan Padhye. SoCC 2021, pp. 388-402.

Contribution: Created Filibuster and its microservice fault-injection tooling. Software.

7. **Durable Functions: Semantics for Stateful Serverless**

Sebastian Burckhardt, Chris Gillum, David Justo, Konstantinos Kallas, Connor McMahon, **Christopher S. Meiklejohn**. Proceedings of the ACM on Programming Languages 5 (OOPSLA), pp. 1-27 (2021).

Contribution: Co-developed the stateful-serverless model and ReactiveMachine prototype that led to Durable Entities.

8. **Composing and Decomposing Op-Based CRDTs with Semidirect Products**

Matthew Weidner, Heather Miller, **Christopher Meiklejohn**. Proceedings of the ACM on Programming Languages 4 (ICFP), pp. 1-27 (2020).

9. **A.M.B.R.O.S.I.A: Providing Performant Virtual Resiliency for Distributed Applications**

Jonathan Goldstein, Ahmed Abdelhamid, Mike Barnett, Sebastian Burckhardt, Badrish Chandramouli, Darren Gehring, Niel Lebeck, **Christopher Meiklejohn**, Umar Farooq Minhas, Ryan Newton, Rahee Ghosh Peshawaria, Tal Zaccai, Irene Zhang. Proceedings of the VLDB Endowment 13(5), pp. 588-601 (2020).

Contribution: Co-developed JavaScript bindings; integrated AMBROSIA prototype components into the main project.

PUBLICATIONS (CONTINUED)

Journal and Conference Papers (continued)

10. **PARTISAN: Scaling the Distributed Actor Runtime**

Christopher S. Meiklejohn, Heather Miller, Peter Alvaro. USENIX ATC 2019, pp. 63-76.

Contribution: Designed and implemented Partisan's Erlang distribution layer and configurable communication mechanisms. Software.

11. **Practical Evaluation of the Lasp Programming Model at Large Scale: An Experience Report**

Christopher S. Meiklejohn, Vitor Enes, Junghun Yoo, Carlos Baquero, Peter Van Roy, Annette Bieniusa. PPDP 2017, pp. 109-114.

Contribution: Built and evaluated the Erlang prototype across 1,024 AWS nodes.

12. **Lasp: A Language for Distributed, Coordination-Free Programming**

Christopher Meiklejohn, Peter Van Roy. PPDP 2015, pp. 184-195. Recipient of the PPDP 2025 10 Year Most Influential Paper Award.

Contribution: Co-designed Lasp with Peter Van Roy and implemented the Erlang prototype. Software.

Demonstration and Poster Papers

13. **Can My Microservice Tolerate an Unreliable Database? Resilience Testing with Fault Injection and Visualization**

Michael Assad, **Christopher S. Meiklejohn**, Heather Miller, Stephan Krusche. ICSE Companion 2024, Demonstrations, pp. 54-58.

Contribution: Mentored Michael Assad's Filibuster research on developer tooling and database resilience testing.

14. **Augmented Inverted Indexes to Track Causality in Eventually Consistent Data Stores**

Christopher Meiklejohn. *Sole author.* Middleware Posters and Demos 2014, pp. 7-8. Poster paper.

Workshop Papers

15. **Composing and Decomposing Op-Based CRDTs with Semidirect Products: (Summary)**

Matthew Weidner, Heather Miller, **Christopher Meiklejohn**. PaPoC 2020, pp. 1-2. Two-page workshop summary; full ICFP 2020 paper listed above.

16. **Partisan: Enabling Real-World Protocol Evaluation**

Christopher S. Meiklejohn. *Sole author.* APPLIED 2018, pp. 45-48.

17. **Towards a Solution to the Red Wedding Problem**

Christopher Meiklejohn, Heather Miller, Zeeshan Lakhani. HotEdge 2018.

18. **On the Design of Distributed Programming Models**

Christopher S. Meiklejohn. *Sole author.* PMLDC at ECOOP 2017, pp. 1-6. Related preprint: arXiv:1701.07615 (2017).

19. **A Certain Tendency of the Database Community**

Christopher S. Meiklejohn. *Sole author.* Salon des Refusés, Programming Companion 2017, pp. 1-5. Earlier preprint: arXiv:1510.08473 (2015).

20. **Loquat: A Framework for Large-Scale Actor Communication on Edge Networks**

Christopher S. Meiklejohn, Peter Van Roy. SmartEdge, PerCom Workshops 2017, pp. 563-568.

21. **Worlds of Events: Deduction with Partial Knowledge about Causality**

Seyed Hossein Haeri, Peter Van Roy, Carlos Baquero, **Christopher Meiklejohn**. ICE 2016, EPTCS 223, pp. 113-127.

PUBLICATIONS (CONTINUED)

Workshop Papers (continued)

22. Towards Property-Based Consistency Verification
Paolo Viotti, **Christopher Meiklejohn**, Marko Vukolić. PaPoC 2016, pp. 1-4.

23. Declarative, Sliding Window Aggregations for Computations at the Edge
Christopher Meiklejohn, Seyed H. Haeri, Peter Van Roy. EdgeCom, IEEE CCNC 2016, pp. 32-37.

24. Selective Hearing: An Approach to Distributed, Eventually Consistent Edge Computation
Christopher Meiklejohn, Peter Van Roy. W-PSDS, SRDS Workshops 2015, pp. 62-67.

25. The Implementation and Use of a Generic Dataflow Behaviour in Erlang
Christopher Meiklejohn, Peter Van Roy. ACM SIGPLAN Erlang Workshop 2015, pp. 39-45.

26. Lasp: A Language for Distributed, Eventually Consistent Computations with CRDTs
Christopher Meiklejohn, Peter Van Roy. PaPoC 2015, pp. 1-4.

27. Derflow: Distributed Deterministic Dataflow Programming for Erlang
Manuel Bravo, Zhongmiao Li, Peter Van Roy, **Christopher Meiklejohn**. ACM SIGPLAN Erlang Workshop 2014, pp. 51-60.

28. On the Composability of the Riak DT Map: Expanding from Embedded to Multi-Key Structures
Christopher Meiklejohn. *Sole author.* PaPEC 2014, pp. 1-2.

29. Riak DT Map: A Composable, Convergent Replicated Dictionary
Russell Brown, Sean Cribbs, **Christopher Meiklejohn**, Sam Elliott. PaPEC 2014, p. 1.

30. Riak PG: Distributed Process Groups on Dynamo-Style Distributed Storage
Christopher Meiklejohn. *Sole author.* ACM SIGPLAN Erlang Workshop 2013, pp. 27-32.

Workshop Manuscripts and Preprints

31. Collabs: Composable Collaborative Data Structures
Matthew Weidner, Heather Miller, Huairui Qi, Maxime Kjaer, Ria Pradeep, Benito Geordie, **Christopher Meiklejohn**. arXiv:2212.02618v1 (2022). Historical preprint, version 1 (2022).

32. Distributed Execution Indexing
Christopher S. Meiklejohn, Rohan Padhye, Heather Miller. arXiv:2209.08740 (2022). Work-in-progress report.

33. Towards A Systems Approach To Distributed Programming
Christopher S. Meiklejohn, Peter Van Roy. arXiv:1802.02681 (2018). Workshop manuscript presented at OBT 2018; arXiv version cited.

34. Partisan: Enabling Cloud-Scale Erlang Applications
Christopher Meiklejohn, Heather Miller. arXiv:1802.02652 (2018). Earlier Partisan preprint; related ATC 2019 paper listed above.

35. Just-Right Consistency: Reconciling Availability and Safety
Marc Shapiro, Annette Bieniusa, Nuno Preguiça, Valter Balegas, **Christopher Meiklejohn**. arXiv:1801.06340 (2018).

36. LightKone: Towards General Purpose Computations on the Edge
Ali Shoker, João Leitão, Peter Van Roy, **Christopher Meiklejohn**. H2020 LightKone project report, December 2016. Non-refereed project report; manuscript dated December 21, 2016.

PUBLICATIONS (CONTINUED)

Workshop Manuscripts and Preprints (continued)

37. Dynamic Path Contraction for Distributed, Dynamic Dataflow Languages

Borja Arnau de Régil Basáñez, **Christopher S. Meiklejohn**. arXiv:1609.01068 (2016). Work-in-progress report; also presented at AGERE 2016.

38. Declarative, Secure, Convergent Edge Computation

Christopher Meiklejohn. *Sole author.* arXiv:1512.04898 (2015). Preliminary report (December 2015); also presented at OBT 2016.

39. Vector Clocks in Coq: An Experience Report

Christopher Meiklejohn. *Sole author.* arXiv:1406.4291 (2014).

40. Eventual Consistency and Deterministic Dataflow Programming: A Case Study of Integrating Derflow with the Riak Data Store

Christopher S. Meiklejohn. *Sole author.* LADIS 2014 workshop manuscript. Author-hosted manuscript.

Published Talk Abstract

41. Lasp: A Model For Distributed, Convergent, Edge Computation

Chris Meiklejohn. *Sole author.* Applicative 2016, p. 40. Published one-page talk abstract.

RESEARCH COLLABORATIONS & FUNDING

SyncFree - CRDT research and industrial collaboration (2013 - 2016 (project))

Researched CRDTs at Basho and Lasp at UC Louvain; linked Riak engineering to large-scale tests of a Rovio-inspired advertisement counter. Consortium industrial partners: Basho, Trifork, Rovio, and Erlang Solutions.

Mesosphere - Lasp at cloud scale (2016 - 2017 research; 2016 consulting)

Evaluated Lasp across 1,024 AWS nodes using Mesos/Marathon while contracting at Mesosphere. Rebuilt Lasp's distribution layer for these experiments, creating Partisan; published the evaluation at PPDP 2017.

LightKone - successful Horizon 2020 proposal (2016 proposal; 2017 - 2019 project)

Contributed to the successful EUR 3.57M general-purpose edge-computing proposal: drafted and revised the startup plan, integrated work-package tasks and deliverables, and incorporated coordinator feedback. Lasp formed part of its technical foundation.

TEACHING & GUEST LECTURES

Carnegie Mellon University - Course Instruction

17-313: Foundations of Software Engineering Co-instructor: Fall 2020; teaching assistant: Fall 2019.

Taught with Claire Le Goues and Michael Hilton; led all recitations across four in-person, remote, and hybrid sections in Fall 2020.

Software Engineering for ML/AI (Fall 2020). Data preparation, model training and evaluation, deployment tradeoffs, production monitoring, and managing model errors and updates.

Microservices (Oct 15, 2020). Service boundaries, independent deployment, distributed-system tradeoffs, and serverless computing; a monolith-decomposition exercise.

DevOps (Oct 20, 2020). Developer ownership and automated delivery pipelines: builds, testing, deployment, monitoring, and continuous integration.

TEACHING (CONTINUED)

17-313: Foundations of Software Engineering (continued)

Chaos Engineering (Oct 29, 2020). Latency, partial failure, game days, and failure experiments: exposing hidden dependencies and improving system and team resilience.

Code Archaeology demonstration (Fall 2020). Live-coded a Rails feature to demonstrate navigating and modifying an unfamiliar codebase.

Chaos Engineering (Fall 2019). Failure experimentation, paired with a practical Gremlin recitation and an industry guest lecture by Tammy Butow.

Carnegie Mellon University - Guest Lectures

Teach guest lectures on software engineering, microservice reliability, and AI-assisted development.

17-313: Foundations of Software Engineering Guest lectures in Michael Hilton's course.

Building Real Software with AI (Mar 31, 2026). Lessons from building Zabriskie with coding agents: debugging, end-to-end tests, CI, device validation, and automated releases.

Deployment (Mar 26, 2026). Safe production deployment and the challenges of releasing changes to live software.

Release lecture series: rollout planning, compatibility, resilience testing, and failure handling.

Reliably Releasing Software (Nov 13, 2025). Feature flags, rollout metrics, schema compatibility, and partial failure; hard versus soft dependencies and fallback testing.

Reliably Releasing Software, Lecture 2 (Apr 10, 2025). Service-level testing, mock drift, and microservice compatibility under partial failure; identifying hard dependencies and validating fallbacks.

Reliably Releasing Software, Lecture 1 (Apr 8, 2025). Rolling deployments, dark launches, database/API compatibility, and tiered deprecation; rollout-metric and rollback-plan exercises.

Reliably Releasing Software (Nov 19, 2024). Feature flags, compatible database migrations, service tests, and mock-drift detection; rollout-planning exercises.

17-712: Fantastic Bugs and How to Find Them Guest lectures in Rohan Padhye's course.

Microservice Testing (Feb 27, 2025). Finding resilience bugs in production microservices through service-level fault injection with Filibuster.

Microservice Resilience (Mar 30, 2023). Systematic fault injection with Filibuster, circuit-breaker behavior, and resilience testing at DoorDash scale.

17-636: DevOps Scheduled guest lectures, Fall 2026.

Reliability and Fault Injection (Nov 2, 2026). Systematic exploration of service failures and resilience testing, drawing on Filibuster.

Harness Engineering (Nov 18, 2026). Tests, feedback loops, and controls for evaluating coding-agent changes and shipping software safely.

TEACHING (CONTINUED)

UC Louvain - Teaching Assistant, 2016 - 2017

LSINF2345: Languages and Algorithms for Distributed Applications (Spring 2017)

Formal models and distributed algorithms; collaborative applications under concurrency and partial failure.

LINGI2145: Cloud Computing (Fall 2016)

French-language course on cloud architecture, scaling, programming models, and large-scale computation.

MENTORSHIP

Michael Issac Assad (Summer 2023) Research Experiences for Undergraduates, Carnegie Mellon University

Mentored research and development on Filibuster, including developer tooling and database resilience testing. Subsequently admitted to the CMU Ph.D. program; co-authored the ICSE 2024 tool demonstration. Related publication [13]

Jacob Wetzel (Summer 2023) Research Experiences for Undergraduates, Carnegie Mellon University

Composable Systems Lab.

Lydia Stark (Summer 2022) Research Experiences for Undergraduates, Carnegie Mellon University

Mentored research on testing microservice circuit breakers with Filibuster; co-authored Method Overloading the Circuit at SoCC 2022. Related publication [4]

Borja Arnau de Régil Basáñez (Summer 2016) Google Summer of Code, BEAM Community / Lasp

Co-mentored a distributed dataflow optimization project applying deforestation through dynamic path contraction. The prototype led to a joint AGERE 2016 paper. Related publication [37]

Georges Younes (Summer 2016) Google Summer of Code, BEAM Community / Lasp

Mentored the project Implementing Operation-Based CRDTs in Lasp, extending the distributed programming model's replicated data structures. INESC TEC project announcement

ACADEMIC & PROFESSIONAL SERVICE

Program committees

ACM SoCC 2024; PaPoC 2021 and 2017; SE-Config 2019; PPDP 2019; Erlang Workshop 2019 and 2015; DEBS 2015.

Workshop founding and leadership

Co-founded and co-chaired Programming Models and Languages for Distributed Computing (PMLDC), co-located with ECOOP, 2016 and 2017, with Heather Miller.

Conference organizing

Curry On organizing committee, 2015 - 2019, including co-chair responsibilities in 2018. Code Mesh co-chair, 2016.

Artifact evaluation

SOSP 2019 artifact evaluation committee.

Additional reviewing

EuroSys 2017 shadow program committee; CloudCom 2016 external reviewer.

Open-source mentorship

BEAM Community, Google Summer of Code 2016: mentored two Lasp projects (see Mentorship).

AWARDS & RESEARCH SUPPORT

PPDP 10 Year Most Influential Paper Award (2025)

With Peter Van Roy, for Lasp: A Language for Distributed, Coordination-Free Programming (PPDP 2015).

MOBILITY Fellowship, Université catholique de Louvain (2016)

Erasmus Mundus Joint Doctorate Fellowship (2015)

Distributed Computing joint doctoral program.

Academic Excellence Award, Northeastern University (2009)

Doctoral research also received support from a Booz Allen Hamilton award and a Software Engineering Education Fund fellowship, as acknowledged in the dissertation.

RESEARCH SEMINARS & VISITS

NII Shonan Seminar 143 - Programming Language Support for Data-intensive Applications (Jul 1 - 5, 2019)

Shonan Village Center, Japan. Presented Equivalence Partitioned Fault Injection; participated in the working group on verification under weak consistency and replicated data types.

Dagstuhl Seminar 18091 - Data Consistency in Distributed Systems: Algorithms, Programs, and Databases (Feb 25 - Mar 2, 2018)

Dagstuhl, Germany. Presented Homomorphic Computation for Distributed Computing, comparing Lasp and BloomL programming models.

Research visits and collaboration (2014; 2016 - 2017)

Research visits in 2014 to Kaiserslautern, Louvain-la-Neuve, Braga, Paris, and London. During 2016 - 2017, visited groups at Cambridge, Kent, Minho, Oxford, and IMDEA.

SUMMER SCHOOLS & FORUMS

Programming Language Implementation Summer School (PLISS) - Bertinoro, Italy (2017)

Heidelberg Laureate Forum (2016)

ECOOP Summer School (2013)

Oregon Programming Languages Summer School (OPLSS) - University of Oregon (2013)

TALKS & WORKSHOPS

Invited Keynote

Distributed Erlang: From Datacenter Applications to Planetary Scale Applications. Erlang Workshop 2018 (keynote).

Research Presentations

Method Overloading the Circuit. ACM SoCC 2022.

Service-Level Fault Injection Testing. ACM SoCC 2021.

C8DB: Geo-Replicated, Conflict-Free Document Database. High Performance Transaction Systems (HPTS), Nov 5, 2019. Macrometa joint work with Durga Gokina and Chetan Venkatesh: geo-replication, transaction semantics, and conflict resolution.

TALKS & WORKSHOPS (CONTINUED)

Research Presentations (continued)

PARTISAN: Scaling the Distributed Actor Runtime. USENIX ATC 2019.

Towards a Solution to the Red Wedding Problem. HotEdge 2018.

Towards a Systems Approach to Distributed Programming. OBT 2018.

Practical Evaluation of the Lasp Programming Model at Scale. PPDP 2017.

A Certain Tendency of the Database Community. Salon des Refusés, <Programming> 2017.

Declarative, Convergent Edge Computation. Northeastern University PL Seminar, October 27, 2016.

Declarative, Sliding Window Computations at the Edge. EdgeCom 2016.

Selective Hearing: An Approach to Distributed, Eventually Consistent Edge Computation. W-PSDS 2015.

The Implementation and Use of a Generic Dataflow Behaviour in Erlang. Erlang Workshop 2015.

Lasp: A Language for Distributed, Coordination-Free Programming. PPDP 2015.

Lasp: Distributed, Eventually Consistent Computations. PaPoC 2015.

Derflow: Distributed Deterministic Dataflow Programming for Erlang. Erlang Workshop 2014.

Eventual Consistency and Deterministic Dataflow Programming. LADIS 2014.

On the Composability of the Riak DT Map: Expanding from Embedded to Multi-Key Structures. PaPEC 2014.

Riak PG: Distributed Process Groups on Dynamo-Style Distributed Storage. Erlang Workshop 2013.

Industry Conferences

Resilient Microservices without the Chaos. Strange Loop 2022.

Partisan: High-Performance Large Scale Distributed Erlang. Code BEAM 2019.

Scaling a Startup with a 21st Century Programming Language. Velocity 2017.

Language Support for Cloud-Scale Distributed Systems. BuzzConf 2017.

Just-Right Consistency: Closing The CAP Gap. J on the Beach 2017; Percona Live 2017.

Loquat: A Design For Large-Scale Distributed Applications. Erlang Factory 2017.

Declarative, Convergent Edge Computation. Oredev 2016; CRAFT 2017.

A Brief History Of Distributed Computing: RPC. YOW! 2016; Code Mesh 2016; Oredev 2016. Code Mesh presentation co-presented with Caitie McCaffrey.

Designing & Evaluating a Distributed Computing Language Runtime. MesosCon 2016; Erlang User Conference 2016.

Lasp: A Model For Distributed, Convergent, Edge Computation. Applicative 2016.

Declarative, Secure, Convergent Edge Computation. GOTO Chicago 2016; Erlang Factory 2016.

A Brief History of Chain Replication. QCon 2015.

Declarative, Distributed, Edge Computation. RICON 2015.

Coordination-Free Designs for Mobile Gaming. CodeMesh 2015.

Distributed, Eventually Consistent Computations. Strange Loop 2015; Erlang Factory 2015.

TALKS & WORKSHOPS (CONTINUED)

Industry Conferences (continued)

Coordination-Free Computations. GOTO Chicago 2015.

Functional Web Apps with Webmachine. BOBkonf 2015; LambdaJam 2013. LambdaJam 2013 workshop co-presented with Sean Cribbs.

Introduction to Riak. BOBkonf 2015.

Building Applications with Distributed Erlang. NDC London 2014.

Eventually Consistent Computations. ReactConf 2014.

Eventually Consistent Computation with CRDTs. RICON 2014.

SyncFree: Large-Scale Computation Without Synchronization. RICON 2014. Co-presented with Annette Bieniusa.

Convergent / Divergent. EmberConf 2014.

Verified Vector Clocks. Erlang Factory 2014; RICON West 2013.

Choose Your Own Consistency. Toronto Erlang Factory Lite 2013.

Bloom: A Language For Disorderly Distributed Computing. Wicked Good Ruby 2013.

A Distributed Process Registry on Riak Core. NYC Erlang Factory Lite 2013.

From Ruby to Erlang: An Unexpected Journey. Chicago Erlang Factory Lite 2013.

An Introduction to Functional Reactive Programming. NodePDX 2013.

Distributed Data Structures in Coq. RICON East 2013.

Building Ambitious Web Applications with Ember.JS. FITC 2012.

Workshops & Community Talks

Conflict-Free Replicated Data Types. Code Mesh 2016 (workshop). Hands-on introduction to CRDTs in Erlang and the Antidote data store.

Implementing Location Independent Invocation. Papers We Love, Paris, April 2016.

HyParView: A Membership Protocol for Reliable Gossip-Based Broadcast. Papers We Love, Madrid, May 2016.

A Brief History of Chain Replication. Papers We Love, San Francisco, 2015.

Bloom, CRDTs, and Distributed Computation. Papers We Love, Paris, July 2015 (whiteboard session).

Consistency and Riak. Riak Meetup, Paris, February 2015.

A Note on Distributed Computing. Papers We Love, Boston, 2014.

Choose Your Own Consistency. London Erlang User Group 2014.

Client Side Applications with Ember.js. Hack Harvard 2013.

What Even Is an Erlang?. Wicked Good Ruby 2013 (lightning talk).

An Introduction to ClojureScript. Boston Clojure, December 2012.

Architecting Ember.js Applications. Boston Ember.js Meetup, October 2012.

COMMUNITY & VOLUNTEER ACTIVITIES

Devio.us - Volunteer Systems Administrator (May 2010 - Mar 2012)

Administered an OpenBSD shell service.

Think Distributed - Creator and podcast host (2013 - 2014)

Created a podcast exploring distributed systems and research with practitioners.

Proceedings - Creator (2013)

Launched an invitation-only beta for online distributed-systems reading groups, with video discussion and an archival model, as a spin-off of Think Distributed.

TECHNICAL WRITING

Using Fault Injection Testing to Improve DoorDash Reliability (2022)

Sole-authored DoorDash Engineering article on adapting Filibuster to industrial microservices and testing resilience before production.

Highly Distributed Computations Without Synchronization (2015)

Sole-authored InfoQ article explaining CRDTs and distributed deterministic dataflow through Lasp examples.

Distributed Programming / Programming Models for Distributed Computation (Book manuscript)

Co-author with Heather Miller, Nat Dempkowski, and James Larisch. Open-source textbook manuscript on distributed programming models; unpublished draft.

Technical blog (2013 - present)

Tutorials and essays on distributed systems, CRDTs, Erlang, fault injection, software engineering education, and AI-assisted development. Includes the distributed programming language history series, Filibuster tutorials, and an eight-part survey of multi-agent systems.

RESEARCH SOFTWARE & OPEN SOURCE

Filibuster - Creator (Created 2021)

Designed and built a service-level fault injection framework that extends functional tests to systematically exercise dependency failures before production. Implemented JVM instrumentation for Kotlin/gRPC microservices. Applied at DoorDash to find resilience bugs before production, including testing of DashPass.

Publications (numbered as in CV): [1], [4], [6], [13], [32]

Partisan - Creator and original maintainer (2016 - 2022)

Designed and implemented an alternative distribution layer for Erlang, with configurable network overlays, parallel connections, named channels, and affinitized scheduling. Added message interposition for causal ordering (Ishikawa) and systematic fault injection; built the runtime for large-scale Lasp experiments. Deployed by LeapSight in LO/JACK LATAM systems; adopted by BarrelDB and used in Bondy and PlumDB. Handed maintenance to Alejandro Ramallo at LeapSight in June 2022.

Publications (numbered as in CV): [10], [16], [34]

OPEN SOURCE (CONTINUED)

Lasp - Co-creator with Peter Van Roy (2014 - 2019)

Co-designed and implemented a coordination-free programming model combining deterministic dataflow with CRDTs. Built and evaluated the Erlang prototype at cloud scale, including experiments across 1,024 AWS nodes. Used in the SyncFree and LightKone research projects and as a storage backend for Erleans. The original paper received the PPDP 2025 10 Year Most Influential Paper Award.

Publications (numbered as in CV): [3], [11], [12]

Riak - Team member and contributor (2012 - 2019)

Maintained internals of this Dynamo-inspired distributed database, including multi-datacenter replication and convergent data types. Built stateful QuickCheck models under John Hughes's mentorship and contributed research on replicated data structures and dataflow integration. Partially oversaw Floppystore development in SyncFree; that project became AntidoteDB. Production users included the UK National Health Service and Riot Games' League of Legends.

Publications (numbered as in CV): [28], [29], [30], [39], [40]

RubyGems.org - Committer and maintainer (Jul 2011 - Dec 2015)

Maintained the Ruby community's package-hosting service.

Riak PG - Creator (2013)

Built a distributed process-group implementation on Riak Core, using replicated data structures to support membership under failures and network partitions.

Publications (numbered as in CV): [30]

Derflow - Research collaborator (2014)

Co-developed distributed deterministic dataflow research for Erlang and explored integration with Riak, including failure testing and a test harness.

Publications (numbered as in CV): [27], [40]

ADDITIONAL TECHNICAL SLIDE DECKS

Archived slides; dates refer to the slide records.

Building Multi-Tenant Applications with Riak CS (2012)

A Brief Introduction to Ember.js (2012)

Ember.js State Manager (2012)

A Brief Introduction to Dialyzer and PropEr (2014)

PODCASTS & INTERVIEWS

Code Mesh speaker interview (2016)

Interview on CRDTs, Lasp, and the Antidote tutorial.

Functional Geekery, Episode 32 (2015)

Guest discussion of Lasp, functional programming, and distributed systems.

Food Fight Show: Erlang Eye for the DevOps Guy and Gal (2013)

Guest discussion of Erlang internals and production use with Basho colleagues.